



METHODOLOGY Creating a Product ECO

STELLANTIS

Type: TRANSVERSAL

PLM METHODOLOGY

CREATING A PRODUCT ECO

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Reference document

DOCINFO storyboards as links:

http://docinfogroupe.inetpsa.com/ead/doc/document_view.action?ref=20061_13_01661&v=vc

UCG as links:

UCG-1534 - Creating an ECO:

UCG-1538 - Completing the attributes of ECO_1

UCG-1535 - Promoting the ECO to Define Component_1

Ref:

Document:

[20061_17_01640](#)

[20061_16_01347](#)

[20061_17_01982](#)

MEMENTO Linking a Product to an ECO

METHODOLOGY Linking an ECO to an Expected Object

METHODOLOGY Add/Remove ECO under ECR

Flowchart of reference

Link to the flowcharts:

- [Referencing of the flowchart in Wall-EE](#)
FLOWCHART_Creation and officialization of a engine and management reference in the PLM :
https://docinfogroupe.stellantis.com/ead/doc/ref.20061_19_00477/v.dp/pi
- Logigramme PLM - Officialisation pièces_JV :
https://docinfogroupe.stellantis.com/ead/doc/ref.20061_18_01522/v.dp/pi
- 20061_16_01236_FLOWCHART_Creating_a_Realized_Raw_Product :
https://docinfogroupe.stellantis.com/ead/doc/ref.20061_17_01924/v.dp/pi
- 20061_17_01708_FLOWCHART_Change_Management_of_a_Defined_Product :
https://docinfogroupe.stellantis.com/ead/doc/ref.20061_17_01956/v.dp/pi
- Flowchart_PLM_process_to_prepare_product_design :
https://docinfogroupe.stellantis.com/ead/doc/ref.20061_21_01114/v.dp/pi
- CREATING A SINGLE DEFINED PRODUCT (PDEF) OR AS NEEDED :
https://docinfogroupe.stellantis.com/ead/doc/ref.20061_16_00938/v.dp/pi
CREATING A SINGLE REALIZED PRODUCT (PREA) :
https://docinfogroupe.stellantis.com/ead/doc/ref.20061_16_00937/v.dp/pi
- LOGIGRAMME_Creation d'un Produit de Définition Brut :
https://docinfogroupe.stellantis.com/ead/doc/ref.20061_15_02819/v.dp/pi
- 20061_15_02355_FLOWCHART_Revising_a_Realized_Product :
https://docinfogroupe.stellantis.com/ead/doc/ref.20061_16_00951/v.dp/pi



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1. Introduction

The purpose of this document is to describe the methodology for creating a Product ECO (engineering change order). The Product ECO is the equivalent of the FM (change sheet) in SAP/3DCom.

2. Principle

2.1. Use case no. 1

First of all, the user should create a Product ECO from the “Action” tool, on the home page. Then, he should connect it to a product, an expected deliverable (LA) or an ECR.

2.2. Use case no. 2

The user has just cloned a defined/realized product in the product library. Then, he associates it to a Product ECO using the “create a Product ECO” function in “linked ECO and ECR” category of the cloned defined/realized product.

2.3. Use case no. 3

The user has just created an ECR, then he associates it to a product ECO using the function “Create a Product ECO” in the category “ECO & ECs” of the “ECR”.

2.4. Prerequisites

- Log-on roles: **PSA_Pilote_Projet / PSA_Concepteur**
- The diversity and applicability contexts are defined
- The Expected Product is created with diversity applicability
- The reviewer and approval lists are created and defined
- The relevant Defined Product (or Realized Product or DOC) is created
- ECR in “**Design**” state in case of revision, or a OA/LA in “Create” state in case of first release or reuse (+ revision for Organ Design Department)



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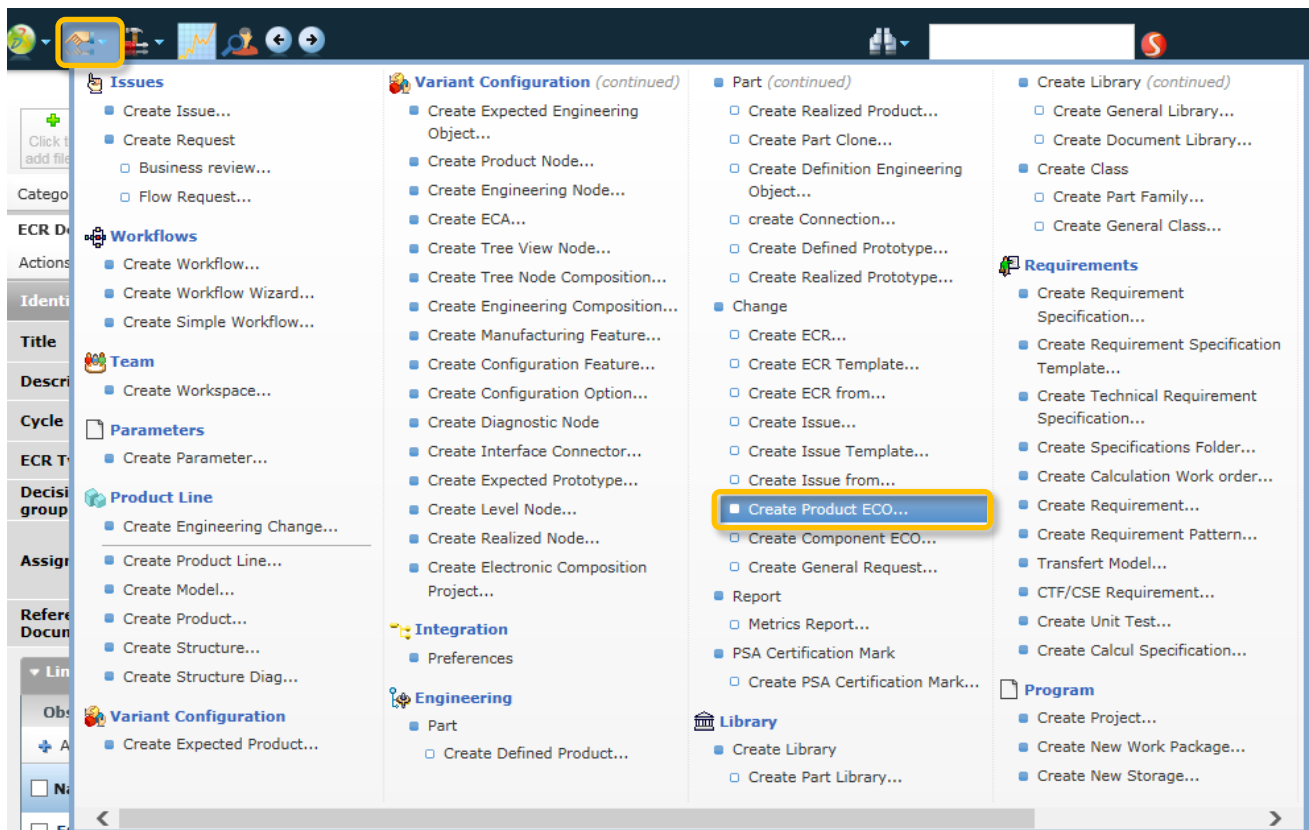
Type: TRANSVERSAL

3. Implementation

3.1. Method 1: Create a Product ECO from PLM Home page, then “Add Existing”

3.1.1. Create a Product ECO

Go to “Actions”, then select the “Create Product ECO” tab



The window for filling the ECO's attributes will open, we should enter the Product ECO attributes



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Fill in all of the fields considered necessary by the designer, then click on “Finish”.

⚠ To be able to release a product, you must fill in the following fields:

- “Description”: Filling in recommendations:
[Electricity/Electronics Link](#)
[Mechanical Engineering Link](#)
- “Due Date”: PROTO or SERIE
- “Homologation Flow”: if the ECO has a PRODUCTION flow (SERIE), the homologation flow must be indicated as “YES”
➡ You must contact the Homologation Manager beforehand to find out if the part concerned by the ECO is indeed subjected to this flow (some PSA breakdowns are exempted from RHN flows and the approver has the list)
“Homologation User”: the ID/name of the approver is required if the “Homologation Flow” indicates “YES”
The Username for homologation is available by link « Homologation User List »
- “Responsible Design Engineer”: enter the name/first name of the component manager (it is required for release)
- “RT Approvers”: enter the Workflow Template for the Technical review RT
- “RAL Approvers”: enter the Workflow Template for the deliverable Approval review RAL
- “CADER”: if the CADER (Vehicle or Component Code) exists for the current release, select it in the list, however, if it does not exist, leave it blank.
- “DUD”
- “Notification’s context”: It is used to alert the notification technician, in order to diffuse the folder to the concerned partners. If the ECO concerns a partner project / JV, it is **imperative** to fill in this field.

Fields in red italics are required

Type	ECO
------	-----

Basic Attributes	
ECO Type	Product
Description	
Maturity	
Homologation Flow	
Notification List	... Clear
Engineering approval	... Clear
Homologation User	... Clear
Homologation User List	http://docinfogroupe.inetpsa.com/ead/doc/ref.01442_11_00131/v.dp/pi
Release approval	... Clear
Design Responsibility	
Responsible Design Engineer	Unassigned ... Clear
Number of ECO to create	1
Identification Description	
Cader	
DUD	
Notification's context	
V/O/M Impacts	... Clear
Manufacturing Site	...



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- “V/O/M Impacts”: the V/O/M impacts can be entered from the ECO (see below)



Warning:

To inform the Impact V / O / M, choose the project space .

Do not choose the Product Line  PL-000000

To fill in the V / O / M impacts field (project space), you must:

1. Check the "Project Space" attribute
2. Search the name of the desired project
3. Check the result (type: project space)
4. Submit.

Search Refinement



Project Type: Project

Taxonomies

Types (Select All | Select None | Expand All | Collapse)

☒ Project Management(10)
☒ Project Space(10)

Attributes

Name

Originated

Modified

COFOR Manufacturer

Context Configuration

Description

Highest

Organization

Originator

Owner

Project

PSA_APQP_MEMBER

Revision

SOD Case

State

Title

Ventilation Context

Ventilation Priority

Search Results

1 - 10 of approx. 10 Results

Page Size: 50 (Max Value: 1000)

Form View Collections Switch to RealTime Publish on CATIA V6 View

Name	Revision	Type	Description	Mastership	PSA Echeance	PSA Candidatu
EB2LInde	27155327075811	Project Space	"EB2 Turbo Gen III Inde Base moteur EB2 Turbo Gen III Europe Decontaining du moteur pour optimisation: suppression des capteurs liés à la norme C7 / passage à des démarreurs et alternateurs conventionnels / passage TGF / suppression thermostat pilote à documenter IL > 90°C Pièces A: choix fournisseurs Monde Pièces C: choix JV avec opportunité d'utiliser les pièces locales en Europe Norme Bharat 6.2 Usine lanceuse: Hosur Gamme GMP : MB6 et AT6 III. STT: avec 1er AMC sur SC24 BVA en 10/2022 "			
MB6	21394112740787	Project Space	Boite de vitesse manuelle 6 rapports (AMC Juillet 2017)			
EB0LTD-FF	901553268399987	Project Space	EB0 Turbo Gen III compatible Flex Fuel (E100) Base moteur EB2 Turbo Gen III Réalisation de la cylindrée par diminution de la course. Evolutions BAP / Vilebrequin / suppression AEB / Poulie AVT / FIE RVC 11.5 Premier développement d'un moteur EB Turbo en FF Norme PL7 + BR3 IL des pièces C (les pièces A seront affecter en amont avec la prise en compte d'un global sourcing incluant l'Amlat), IL des 3C (cible totale: 1CR à 60%) Usine lanceuse UMPR Gamme GMP : MB6 et AT6 III. STT: avec Compatibilité PTF : CMP et SC 1er AMC sur CMP AT6 III en 11/2022 (demande, faisabilité à confirmer)			
CMPP0E63	761544715160872	Project Space	Evolution majeure de la Base CMP pour intégrer l'hybridation P0 sur moteur EB2ADT MB6 Euro6.3 dans les sites de fabrication DEUR+DMOA			
MB6-CH	11492676311767	Project Space	MB6 Chine DPCA			
Migration PLW abandonner	31512144713143	Project Space	Migration PLW			
			Programme de remotorisation Plateforme CMP : Euro7, China7, Sulev30 (Corée) et réduction du CAFE/CAFC (introduction MUSE)			

 Reset

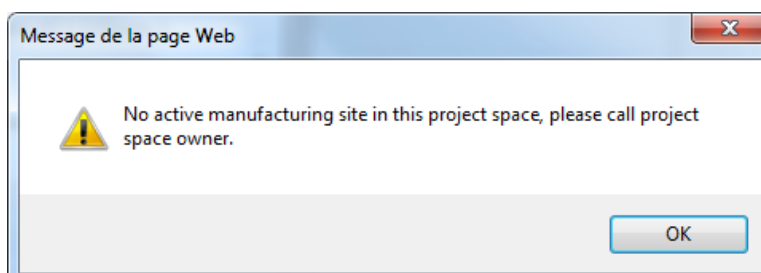








If you have an error message (see below), you must contact the Project Owner (RPP or RPCR) to fill in the manufacturing site(s) on the project space





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5. Choose one or more manufacturing sites

Business Unit



MB6

- ☐ VALENCIENNES
- ☐ ASPERN
- ☐ CHINE CAPSA (SHENZHEN)
- ☐ INDE

These manufacturing sites will be automatically entered in the field "Manufacturing Site"

V/O/M Impacts

MB6

Clear

Manufacturing Site

VALENCIENNES

...

If necessary, it is possible to modify the field by clicking on ...

Note: When only one manufacturing site is associated with a VOM impact, this manufacturing site is automatically displayed by clicking on ...

Note:

- To confirm the creation of the ECO, you must fill in the attributes in red
- The attributes in black can be filled in after creating the ECO, but only if the item has the "Create" or "Define component" maturity.

With this method, several ECOs can be created in one go. Simply enter the number of ECOs to be created.

Design Responsibility	PSA
Responsible Design Engineer	Unassigned ... Clear
Number of ECO to create	1
Identification Description	
Cader	

Example for three ECOs created in one go:

Create Product ECO												
Mass Update												
Name	ECO Type	Description	Maturity	Homologation Flow	Notification List	RT Approvers	Homologation User	RAL Approvers	Design Responsibility	Responsible Design Engineer	Cader	DUD
☐ ECO-0031157	Product	PDEF PROTO	PROTO	No		RT-000311		RAL_TEST	PSA	e541324		1 - 1er Présen
☐ ECO-0031158	Product	PDEF PROTO	PROTO	No		RT-000311		RAL_TEST	PSA	e541324		1 - 1er Présen
☐ ECO-0031159	Product	PDEF PROTO	PROTO	No		RT-000311		RAL_TEST	PSA	e541324		1 - 1er Présen
3 objects												

3.1.2. Add an existing ECO to a PDEF/PREA, DOC, OA/LA, or ECR

Add an existing ECO to a PDEF (or DOC):

[MEMENTO Linking a Product to an ECO](#)

Add an existing ECO to an OA/LA (expected deliverable):

[METHODOLOGY Linking an ECO to an Expected Object](#)

Add an existing ECO to an ECR:

[METHODOLOGY Add/Remove ECO under ECR](#)



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3.2. Method 2: Create a product ECO from the identity card of a PDEF/PREA, or a DOC

Click to add files

Add to clipboard Add to Collection

Refresh view

Defined Product A-00071512-00 (1) PSA D34E

Originator: HARY Rudy

Categories

Actions 3D 3DLive Examine Create PSA reference Open the Report

FAISCEAU+PLANCHE+BORD+DA

G

- Affected Product Lines
- Alts/Subs
- APV Tooling
- Au Titre De
- Classification
- Costs
- Derived Parts
- Distribution
- Equivalents
- Functional
- GB of Structure
- History
- Issues & Requests
- Lifecycle
- Linked Action Plan
- Linked ECO and ECR**
- Logical
- Logical Structures
- Low Interchangeability

FAISCEAU+PLANCHE+BORD+DA

G (suite)

- Ownership Details View
- Partners References
- Physical
- Product Composition Management
- PSA Rapport
- PSA References
- Realized products
- Reference Documents
- Referenced By
- Revisions
- Spare Parts
- Specifications Documents
- Where Used In MBOM
- Workflow

- Open « Categories » tab
- Then, click on « [Linked ECO and ECR](#) »

As the « [Linked ECO and ECR](#) » window opens:

Click to add files

Add to clipboard Add to Collection

Refresh view

Defined Product A-00071512-00 (1) PSA D34E

Originator: HARY Rudy State: In Work

Originated: 12/09/2015 Validate

Owner: HARY Rudy Modified: 09/22/2016

Linked ECO and ECR

Categories ?

ECRs **ECOs**

Add Existing Product.. **ECO Create Product ECO** Remove View

Name	Type	Description	Disp Codes
No Objects Found			

- Select the “ECO” sub-tab
- Then, click on “[Add Existing Product ECO](#)”

The Product ECO creation window is displayed on the « [Linked ECO and ECR](#) » view.

The advantage of this method, is to create an ECO and to link it directly to a product or a doc, at the same time.

When the ECO information window is displayed, we can fill in the fields using the method shown in [paragraph 3.1.1](#)



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3.3. Use case no. 3: Create a Product ECO under the ECR

The ECR has to be at status “Design”.

Go to the identity card of the ECR where we want to create the ECO, then to categories, click on “ECOs & ECs”.

The screenshot shows the identity card for ECR-0000123. The 'Categories' dropdown menu is open, and 'ECOs & ECs' is selected. The 'Properties' section on the right shows the ECR details: Originator: LECHARTIER CHRISTOPHE, State: Closed, Originated: 02/09/2016, Owner: LECHARTIER CHRISTOPHE, Modified: 10/25/2018. The 'Properties' table lists various attributes: Current Duration (days), Due Date, Rescheduling proposal, Solution type, and Business leader.

Current Duration (days)	Due Date	Rescheduling proposal	Solution type	Business leader
			Final solution	DRD/DCHM/CCHE/SPRO

In the **Structure View** of the ECR, go to “Actions”, then click on “create Product ECO...”

The screenshot shows the Structure View for ECR-0021402. The 'Actions' dropdown menu is open, and 'Create Product ECO...' is selected. The 'Structure View' table lists various components: Name, State, and Action. The 'Create Product ECO...' action is highlighted.

Name	State	Action
ECO-0042	Define Components	Create Product ECO...

When the ECO creation window appears, fill in the fields as mentioned in the [3.1.1 paragraph](#)

Note: if a colored products are associated to the product under the PA, also see [4.3 paragraph](#)



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4. Typical cases

4.1. ECO date

For Managing purposes, we can add a due date to the ECO.

From the identity card of the ECO, or via the category “Lifecycle” of the ECO, go to “Approvals” tab.

- From the Identity Card :

ECO-0000123 : Home

Click to add files

Add to clipboard Add to Collection Refresh view

ECO ECO-0000123 Elec Ingenierie TEST MINI OLG

Originator : LEGALL Olivier Originated : 02/28/2014 Owner : LEGALL Olivier

State : Create 0 Specifications Documents Modified : 08/13/2019

Properties

Categories Cost Summary Report Summary Report Add To Folder

Notification List

Design Responsibility Elec

Responsible Design Engineer Unassigned

Diversity Manager Unassigned

Identification Description

Officialization Number AAAAAA00000123

Cader

DUD 1 - 1er Véhicule Présérie

Notification's context

Affected Items Resolved Items Implemented Items Impact Ana Approvals Assignees Reference Documents Loading....

Add Approver Add a PSA Approver Add From Template Edit View

State	Assignee	Task/Signature	Task Title	Comments/Instructions	Action	Due Date	Completed Date
-------	----------	----------------	------------	-----------------------	--------	----------	----------------

From the category “Lifecycle”:

ECO-0000123 : Lifecycle

Click to add files

Add to clipboard Add to Collection Refresh view

ECO ECO-0000123 Elec Ingenierie TEST MINI OLG

Originator : LEGALL Olivier Originated : 02/28/2014 Owner : LEGALL Olivier

State : Create 0 Specifications Documents Modified : 08/13/2019

Lifecycle/Review PowerView

Categories Lifecycle

ECO-0000123 rev : Lifecycle

Validate Promote Demote

Create Define Co... Wait App. Reviewed Released Implement... Cancelled

Tasks/Signatures Approvals

Add Approver Add a PSA Approver Add From Template Edit View

State	Assignee	Task/Signature	Task Title	Comments/Instructions	Action	Due Date	Completed Date
☐		Create					
☐		Define Components					
☐		Wait Ann.					

Click on “Edit”:

Tasks/Signatures Approvals

Add Approver Add a PSA Approver Add From Template Edit View

State	Assignee	Task/Signature	Task Title	Comments/Instructions	Action	Due Date	Completed Date
☐		Create					
☐		Define Components					



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This tab opens, and it allows to edit the due date:

Approvals

Column: Due Date Value (dd/MM/yy) 24/11/19 Apply to Selected Apply to all

	Assignee	Task/Signature	Task Title	Comments/Instructions	Action	Due Date (dd/MM/yy)	Completed Date (dd/MM/yy)
☐	Define Components						
☒	Wait App. BUREL, HUBERT		Valider la RT_DCHM_CMP	Valideur Secondaire	Pending		
☒	Wait App. DUCLOS, PIERRE		Valider la RT_DCHM_CMP	Valideur Principal	Pending		
☐	Reviewed SALOU, MAURICE		Valider RAL	Valider RAL	Pending		
☐	Reviewed BIDAULT, FLORENCE		Valider RAL	Valider RAL	Pending		
☐	Reviewed PLANTEVIN, ANTOINE		Valider RAL	Valider RAL	Pending		
☐	Reviewed HUGUENIN, ROLAND		POUR INFO	Officialisations CMP à répartir	Pending		
☐	Released BEN BERKAT, ABDEL						
☐	Released SALESSE, ANTOINE						
☐	Released PLANTEVIN, ANTOINE						
☐	Released MONCEAU, STEPHANE						
☐	Released RD2/DPVS/PMO/PMO/SOP2						
☐	Released BIDAULT, EDOUARD						
☐	Implemented						
☐	Cancelled						

Calendar: November 2019

Assignee: BUREL, HUBERT

Task/Signature: DUCLOS, PIERRE

Task Title: Valider la RT_DCHM_CMP

Comments/Instructions: Valideur Secondaire

Action: Pending

Due Date: 24/11/19

Completed Date:

Buttons: Done, Cancel

Click on the calendar (for RT workflow approval or RAL), choose a date, then click on apply to selected, then on done.

We now have a due date on the needed tasks related to the concerned Workflow.

Affected Items							
Resolved Items							
Implemented Items							
Impact Analysis							
Approvals							
Assignees							
Reference Documents							
Add Approver Add a PSA Approver Add From Template Edit View							
State	Assignee	Task/Signature	Task Title	Comments/Instructions	Action	Due Date	
Create							
Define Components							
Wait App.	BATS, Romain - RD2/DAPF/MTSV/VDE/COVP	IT-0000100	Workflow de RT (revue)	Test mini en prod : Bon pour passage en RAL	Approved	03/01/2014	
Reviewed	BATS, Romain - RD2/DAPF/MTSV/VDE/COVP	IT-0000101	Workflow de RAL (approbation) - u358489	Test mini pour prod : bon pour offi	Approved	03/01/2014	
Implemented							
Cancelled							

4.2. ECO used for the Creation of a Pdef « Neutral»

When it is necessary to define colors, it is necessary to create a "Neutral" Pdef.

In order for the diversity manager to be informed of the creation of a "Neutral", it is necessary to enter his ID on the ECO that will be used.

⇒ Go to the ECO identity card and click on Actions / Edit details

Categories: Actions Add To Folder

Reference Documents: Edit Details Cost Summary Report Summary Report

Maturity:

Homologation view:

Engineering approval: RT_D34E

Homologation User:

Release approval: Approuver RAL D31E

Notification List:

Design Responsibility: Carrosserie

Responsible Design Engineer: KANTER PATRICIA

Diversity Manager: Unassigned

Use the search engine to find and submit the ID of the diversity manager.

Diversity Manager: Unassigned Clear



4.3. ECO for revision / for reuse of a PDEF instantiated under a PA containing colored products

The screenshot displays the PLM interface for managing ECOs. At the top, there are navigation buttons like 'Add to clipboard', 'Add to Collection', and 'Refresh view'. The main table lists ECOs with columns: Name, Revision, Type, State, Qty, Substance Qty, Configuration Context, Status Relationship, GBOM Link Modification Responsibility, and Quantity Modification Responsibility. Two ECOs are highlighted: A-00181070-00 (Revision 1, Defined Product, Released) and COL-00027964 (Revision 2, Defined Product, Release). Below the table, a detailed view of an ECO is shown, including fields for Assignees, Reference Documents, Maturity, Homologation Flow, Engineering approval, Homologation User, Homologation User List, Release approval, Notification List, Responsible Design Engineer, and Diversity Manager. The Diversity Manager field is currently 'Unassigned'. A message box on the right states: 'Promotion Failed: There are Neutral products connected to the ECO, you must enter a Diversity Manager ID.'

When we use an ECO to modify a PA structure, containing a PDEF and colored products, we have many specific cases, with or without changing the PSA reference of the new PDEF we attached under the PA:

- Evolution 1: We revise the PDEF. The PSA reference doesn't change.
→ There is no need to alert the RCD
- Evolution 2: We create a non-interchangeable revision of the PDEF. The PSA reference changes.
→ We need to alert the RCD
- Evolution 3: We replace the PDEF by another. The PSA reference changes.
→ We need to alert the RCD

For cases no 2 and 3, it is necessary to verify if there is a diversity manager ID in the ECO. He received an alert email, then. If there isn't, we will have the system blocked.